TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	David Koch	**Email**	David.Koch@ipa.fraunhofer.de
Institute Name	Fraunhofer Institute for Manufacturing Engineering and Automation IPA			
Address	Nobelstr.12, 70565 Stuttgart Germany			

2. Technology Overview

Title of Proposed Technology	Smart Grids and improved Energy Systems and Efficiency			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (), Next-gen Nuclear Power (), Advanced Bio (), Aerospace/Marine (), Hydrogen (), Cybersecurity (), AI (X), Next-gen Communication (), Advanced Robotics & Manufacturing (X), Quantum ()			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Smart Grid	Industrial energy management	Energy efficiency improvement	common foundation
Concept of Proposed Technology	Improved energy efficiency using intelligent energy management algorithms incl. AI features and optimization of manufacturing energy systems			
Rationale for Joint R&D Project	Transfer of Research Results into practical application.			
Key Words	1. intelligent data analysis and optimization 2. flexibility			

	3. matching of energy supply and utilization 4. optimized microgrids

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level 4	**End of the Project**	Level 6
R&D Objectives	Demonstrate a scalable industrial AC or DC smart grid with forecast-driven demand-side management and procurement optimization in a pilot production line; quantify energy cost savings and emissions reductions			
R&D Target Specifications	- DC distribution and microgrid design for industrial facilities - AI-based forecasting for energy demand and energy prices - Flexibility services (demand response) for on-site load shifting - Procurement optimization for energy supply contracts - Integration with heating/thermal energy systems (heat pumps) for energy efficiency - Data security and cyber-physical security considerations - Validation in a real or simulated production line			
R&D Plan	Analysis of status quo of production, identification of energy efficiency potentials, matching of energy supply and consumption, sizing and matching of potential energy storage measures, grid optimization (e.g. potential of DC grids).			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Research lead and provider of knowhow - Optimization using proprietary technology and algorithms
Expected R&R of Korean SME Partner	- Provide data and access to information required to perform the analysis - Implementation of identified measures

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Energy cost reductions (target % or absolute) - Improved energy efficiency in production - Increased share of on-site renewables and/or on-site energy storage - Enhanced grid resilience and reliability - Transferable knowledge and potential for new services/products - Jobs and capability development

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization